

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-126740891})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}415 + \mathbf{Z}(176 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 31685223 = 0$, of discriminant -126740891 . Class group invariants $(195 \quad 5 \quad 5)$, class number 4875; the three stored generator ideals (HNF) are $\begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 51)y^3 + (22s - 130606)y^2 + (-73s + 1282752)y + (944s - 8886784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 87y^8 - 260778y^7 + 35298066y^6 - 1408732864y^5 + \\ & 36955715157y^4 - 495711807334y^3 + 5451302849399y^2 - 2716420442 \\ & 0768y + 107202383621888 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375} \right) s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2517218714932 = 2^2 \cdot 629304678733$:

$$\begin{pmatrix} 1258609357466 & 782598244 & 295823383280 & 1040807781051 & 374714318371 & 881523773442 & 446703420967 & 28559662847 & 374797064752 & 1061214248295 \\ 0 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 51)y^3 + (22s - 130606)y^2 + (-73s + 1282752)y + (944s - 8886784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 - 87y^8 - 260778y^7 + 35298066y^6 - 1408732864y^5 + 36955715157y^4 - 495711807334y^3 + 5451302849399y^2 - 27164204420768y + 107202383621888$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201} \right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1770080412983 = 79 \cdot 22406081177$:

$$\begin{pmatrix} 1770080412983 & 1510714998347 & 1120869054289 & 112190521040 & 339361891663 & 1204542237616 & 782737303535 & 236549689332 & 1220294746920 & 361780794306 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 429 factors with signed exponents of absolute value up to 47181946632638440857675912; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 51)y^3 + (22s - 130606)y^2 + (-73s + 1282752)y + (944s - 8886784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 - 87y^8 - 260778y^7 + 35298066y^6 - 1408732864y^5 + 36955715157y^4 - 495711807334y^3 + 5451302849399y^2 - 27164204420768y + 107202383621888$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $68983994758064 = 2^4 \cdot 17 \cdot 19273 \cdot 13159219$:

$$\begin{pmatrix} 8622999344758 & 6781122146730 & 4516605923366 & 5130270560533 & 2352677385784 & 5332181867908 & 2595910204892 & 3898891182244 & 3260057641759 & 4861137617409 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 434 factors with signed exponents of absolute value up to 29768691156393232794117728; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3343y^3 + (-12s + 6)y^2 + 4877203y + (-7872s + 3936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 6686y^8 + 20930055y^6 + 37171651334y^4 + 29773334866921y^2 + 1963482050497536$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375}\right) s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2310192872676 = 2^2 \cdot 3^3 \cdot 13 \cdot 1645436519$:

$$\begin{pmatrix} 385032145446 & 72463898364 & 185662190274 & 247564315267 & 309220926069 & 305524751703 & 246419462003 & 379415088796 & 330071315739 & 26879822676 \\ 0 & 6 & 2 & 0 & 1 & 0 & 0 & 4 & 0 & 3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 916676568633958379; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = \begin{pmatrix} 2 & 0 & 2 \end{pmatrix}$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $\begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3343y^3 + (-12s + 6)y^2 + 4877203y + (-7872s + 3936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 6686y^8 + 20930055y^6 + 37171651334y^4 + 29773334866921y^2 + 1963482050497536$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $\begin{pmatrix} 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201}\right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $52984325659248 = 2^4 \cdot 3^3 \cdot 4639 \cdot 26438651$:

$$\begin{pmatrix} 1471786823868 & 139866380796 & 547896065202 & 242670393129 & 1144414346761 & 731424363391 & 40963888507 & 980917637800 & 385201276751 & 99080988602 \\ 0 & 12 & 0 & 4 & 9 & 2 & 4 & 4 & 4 & 7 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 173 factors with signed exponents of absolute value up to 586528490631369531; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3343y^3 + (-12s + 6)y^2 + 4877203y + (-7872s + 3936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 6686y^8 + 20930055y^6 + 37171651334y^4 + 29773334866921y^2 + 1963482050497536$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $791337809793024 = 2^{12} \cdot 3^3 \cdot 13^2 \cdot 181 \cdot 233923$:

$$\begin{pmatrix} 118890896904 & 106903926584 & 78981328684 & 16017718450 & 16397287810 & 110003149150 & 63538484455 & 19080543696 & 22351785982 & 36465493692 \\ 0 & 104 & 8 & 96 & 26 & 60 & 16 & 96 & 32 & 80 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 1526123369717270750; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 1257)y^3 + (10s + 156160)y^2 + (77s - 563510)y + (-186s - 499926)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 2517y^8 + 307304y^7 + 31512412y^6 - 240007962y^5 \\ & + 21262501393y^4 - 116663455444y^3 + 231321854097y^2 - 3440986085 \\ & 34y + 1346200966620 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375} \right) s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43691997566160 = 2^4 \cdot 3^6 \cdot 5 \cdot 773 \cdot 969181$:

$$\begin{pmatrix} 67425922170 & 18635752350 & 65079686430 & 23704313484 & 57490566268 & 44112431072 & 60551202973 & 19707348334 & 55412101289 & 29530809219 \\ 0 & 18 & 0 & 6 & 8 & 8 & 6 & 5 & 4 & 6 \\ 0 & 0 & 6 & 0 & 5 & 0 & 0 & 4 & 0 & 1 \\ 0 & 0 & 0 & 6 & 5 & 4 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 175 factors with signed exponents of absolute value up to 68363451248169; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 1257)y^3 + (10s + 156160)y^2 + (77s - 563510)y + (-186s - 499926)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 2517y^8 + 307304y^7 + 31512412y^6 - 240007962y^5 \\ & + 21262501393y^4 - 116663455444y^3 + 231321854097y^2 - 3440986085 \\ & 34y + 1346200966620 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201} \right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4738365657936 = 2^4 \cdot 3^5 \cdot 41 \cdot 3529 \cdot 8423$:

$$\begin{pmatrix} 21936878046 & 14301118134 & 4266067230 & 18788553654 & 11090729859 & 21500582275 & 19103428410 & 21895977265 & 13777821995 & 5232960196 \\ 0 & 6 & 0 & 0 & 0 & 0 & 2 & 2 & 4 & 5 \\ 0 & 0 & 6 & 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 184 factors with signed exponents of absolute value up to 121393102922285; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (4 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 1257)y^3 + (10s + 156160)y^2 + (77s - 563510)y + (-186s - 499926)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 2517y^8 + 307304y^7 + 31512412y^6 - 240007962y^5 \\ & + 21262501393y^4 - 116663455444y^3 + 231321854097y^2 - 3440986085 \\ & 34y + 1346200966620 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4738365657936 = 2^4 \cdot 3^5 \cdot 41 \cdot 3529 \cdot 8423$:

$$\begin{pmatrix} 21936878046 & 14301118134 & 4266067230 & 18788553654 & 11090729859 & 21500582275 & 19103428410 & 21895977265 & 13777821995 & 5232960196 \\ 0 & 6 & 0 & 0 & 0 & 0 & 2 & 2 & 4 & 5 \\ 0 & 0 & 6 & 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 185 factors with signed exponents of absolute value up to 66477891663197; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (4 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1448y^3 + (-14s + 7)y^2 - 721792y + (13184s - 6592)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2896y^8 + 653120y^6 + 8300613291y^4 - 11175679657344y^2 + 5507457485287424$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375}\right)s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $41352840998453 = 19391 \cdot 31123 \cdot 68521$:

$$\begin{pmatrix} 41352840998453 & 40977124414409 & 2954119126460 & 36216802919142 & 24950752149833 & 8245624489039 & 31609257452534 & 31425286576557 & 26519198189036 & 9481127578612 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 738 factors with signed exponents of absolute value up to 438725811777816756244842555562531376; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1448y^3 + (-14s + 7)y^2 - 721792y + (13184s - 6592)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2896y^8 + 653120y^6 + 8300613291y^4 - 11175679657344y^2 + 5507457485287424$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201} \right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6861571930844 = 2^2 \cdot 191161 \cdot 8973551$:

$$\begin{pmatrix} 3430785965422 & 2806524789482 & 1362905018369 & 386679145455 & 1398429903459 & 440085774721 & 878042214922 & 2606445249317 & 2443686250258 & 673814074885 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 738 factors with signed exponents of absolute value up to 250641831182934151437263084473360029; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1448y^3 + (-14s + 7)y^2 - 721792y + (13184s - 6592)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2896y^8 + 653120y^6 + 8300613291y^4 - 11175679657344y^2 + 5507457485287424$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $81484156074064 = 2^4 \cdot 519091 \cdot 9810919$:

$$\begin{pmatrix} 10185519509258 & 4610258658268 & 5354354227806 & 6991330801755 & 5591520358224 & 8111059034923 & 5031267787071 & 2395650226613 & 7541295183605 & 5317919191739 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 738 factors with signed exponents of absolute value up to 226667133424010189826279948868396661; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1584y^3 + 184401y^2 + (-332s - 7592554)y + (17296s + 100545784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 3164y^8 + 375138y^7 - 13413988y^6 - 352702624y^5 + 57655248033y^4 - 3118766762016y^3 + 98226544621260y^2 - 1890853628692256y + 19589879383825088$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375}\right) s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $925818819357461 = 71 \cdot 263 \cdot 269 \cdot 184314553$:

$$\begin{pmatrix} 925818819357461 & 357809260895362 & 841987457645357 & 747458974860657 & 202630608866020 & 49494708977954 & 393208806204374 & 33768649003996 & 790702762627633 & 745356448182395 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 376 factors with signed exponents of absolute value up to 2200257927144; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1584y^3 + 184401y^2 + (-332s - 7592554)y + (17296s + 100545784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 3164y^8 + 375138y^7 - 13413988y^6 - 352702624y^5 \\ & + 57655248033y^4 - 3118766762016y^3 + 98226544621260y^2 - 1890853 \\ & 628692256y + 19589879383825088 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201}\right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $67523144841488 = 2^4 \cdot 7^2 \cdot 181 \cdot 647 \cdot 735451$:

$$\begin{pmatrix} 1205770443598 & 675329168994 & 54975896324 & 750477486612 & 329087020088 & 55919660406 & 622745718921 & 1038544992322 & 1041552168861 & 869585195634 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 14 & 0 & 9 & 3 & 11 & 0 & 9 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 376 factors with signed exponents of absolute value up to 2475892142333; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1584y^3 + 184401y^2 + (-332s - 7592554)y + (17296s + 100545784)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 3164y^8 + 375138y^7 - 13413988y^6 - 352702624y^5 \\ & + 57655248033y^4 - 3118766762016y^3 + 98226544621260y^2 - 1890853 \\ & 628692256y + 19589879383825088 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2173202744215724 = 2^2 \cdot 13^2 \cdot 683 \cdot 4706876953$:

$$\begin{pmatrix} 1086601372107862 & 889783596872784 & 178905456239736 & 947155631605467 & 703368614732151 & 741830416383116 & 287360132477867 & 969881359777380 & 1039919908294475 & 493121059999661 \\ 0 & 2 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 376 factors with signed exponents of absolute value up to 2826787881291; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (2 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3371y^3 + (12s - 6)y^2 + 1686955y + (-1956s + 978)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6742y^8 + 14737551y^6 - 6810778534y^4 + 1358386075249y^2 + 121225634387244$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3102493}{12309502009375} + \left(\frac{291}{12309502009375}\right)s, \quad J = \begin{pmatrix} 415 & 176 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 415$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $19095996670704 = 2^4 \cdot 3^3 \cdot 163 \cdot 853 \cdot 317923$:

$$\begin{pmatrix} 795666527946 & 50581444248 & 618726207240 & 413828566666 & 254329067945 & 714243757729 & 565712375551 & 397215881105 & 72194133087 & 549256689123 \\ 0 & 6 & 0 & 4 & 5 & 0 & 2 & 2 & 5 & 5 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 322 factors with signed exponents of absolute value up to 73996438; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3371y^3 + (12s - 6)y^2 + 1686955y + (-1956s + 978)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6742y^8 + 14737551y^6 - 6810778534y^4 + 1358386075249y^2 + 121225634387244$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{219809146}{3255243551009881201} + \left(\frac{318135}{3255243551009881201} \right) s, \quad J = \begin{pmatrix} 5041 & 4727 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5041$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $101681239707324 = 2^2 \cdot 3^6 \cdot 41 \cdot 163 \cdot 5217733$:

$$\begin{pmatrix} 627661973502 & 513565071063 & 32688214848 & 252257413392 & 266483833300 & 403564942337 & 515256888573 & 125387165426 & 574009040640 & 551650777958 \\ 0 & 3 & 0 & 2 & 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 9 & 3 & 8 & 7 & 6 & 1 & 1 & 8 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 308 factors with signed exponents of absolute value up to 66703973; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3371y^3 + (12s - 6)y^2 + 1686955y + (-1956s + 978)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6742y^8 + 14737551y^6 - 6810778534y^4 + 1358386075249y^2 + 121225634387244$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{993784579}{2129972052005212151} + \left(-\frac{189862}{2129972052005212151}\right)s, \quad J = \begin{pmatrix} 4631 & 1002 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4631$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $320462439669756 = 2^2 \cdot 3^8 \cdot 41 \cdot 297826439$:

$$\begin{pmatrix} 659387735946 & 381308619558 & 314026880400 & 515409886578 & 517692919915 & 539071366721 & 239337232470 & 163353702779 & 516330574221 & 248760653069 \\ 0 & 6 & 0 & 0 & 1 & 4 & 0 & 0 & 4 & 1 \\ 0 & 0 & 9 & 0 & 5 & 1 & 6 & 7 & 4 & 5 \\ 0 & 0 & 0 & 3 & 1 & 2 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 333 factors with signed exponents of absolute value up to 1419641881; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.